RK330-01 Communication Protocol（MODBUS-RTU）

1.Communication parameters(factory default):

Baud rate: 9600bps, Data bits:8,Stop bit:1,Check bit:no,Address:0x01

2.Examples for read data

Host Scan Order：

Slave id	Function code	Address_H	Address_L	Quantity_H	Quantity_L	CRC_L	CRC_H
0x01	0x03	0x00	0x00	0x00	0x03	0X05	0XCB

Sensor Response

Slave id	Function code	Number of bytes	Temperature (Unsigned integer 2 bytes)	Humidity (Unsigned integer 2 bytes)	Pressure (Unsigned integer 2 bytes)	CRC_L	CRC_H
0x01	0x03	0x06	0x0121	0x0164	0x2728	0XC7	0X6E

Temperature:(0121)H=(298)D,289/10=28.9℃(If the data≧0x8000, for example:0xFF05,according to the following method to calculate：

0xFF05-0xFFFF-0x01=(65285)D-(65535)D-(1)D=(-251)D,-251/100=-2.51(℃))

Humidity:(0164)H=356(D), 356/10=35.6%

Pressure:(2728)H=10024(D),10024/10=1002.4mbar

3.Examples for modify the address(After the restart to take effect)

Host Scan Order(change 01H to 02H)：

Slave id	Function code	Address_H	Address_L	Reserved	New id	CRC_L	CRC_H
0x01	0x06	0x00	0x00	0x00	0x02	0x08	0x0B

Sensor Response

Slave id	Function code	Data length	Reserved	New id	CRC_L	CRC_H
0x01	0x06	0x02	0x00	0x02	0x39	0X49

Note: If you forget the original address, you should use the broadcast address(FEH) (ensure that no other devices on the bus at this time.

4.Register description

Register name	Function code	Register address	Data type	Data length（byte）	R/W	Remark
Temperature	0x03	0x0000	Signed integer	2	R	℃
Humidity	0x03	0x0001	Unsigned integer	2	R	%
Pressure	0x03	0x0002	Unsigned integer	2	R	kPa
Modify Sensor address	0x06	0x0000	Unsigned integer	1	W	0-0x7F
Baud rate	0x06	0x0080	Unsigned integer	2	W	0x1200:1200 0x2400:2400

						0x4800:4800 0x1150:115200
Parity	0x06	0x0081	Unsigned integer	2	W	0: NONE 1: ODD 2: EVEN